National Science Foundation Grant Terminations Under the Trump Administration

Nicole Basler

College of Liberal Arts, Southern Illinois University

Abstract

This paper will explore and analyze the Dataset about the 2025 National Science Foundation Grant Terminations Under the Trump Administration, a dataset from GitHub in the tidyverse data sets in tidy Tuesday. This data was created by Grant Watch and is continuing to document all the grant terminations that have been going on since 2025, and then this data was transferred into a readable CSV to git hub by Noam Ross and Johnthegeek. This project will focus on identifying which of the NSF Directorates, States, and Public/Private organizations experienced the impacts from these grant terminations. This exploratory data analysis was conducted using R by creating tables, data visualizations, and tests on the data that I used. I had found which states and the type of organizations had the most grant money terminated. This project will also highlight the importance of statistical testing and data visualization for being able to understand the major impact that federal funding has on the changes of research institutions and much more.

Keywords: National Science Foundation(NSF), Grant Terminations, Federal Funding, Data Visualization, education

Introduction

Any terminated grant is more than just cash or a line in a dataset; It can represent the lost research, motivation, education, and benefits from the termination of grants and funding that ultimately disrupted many students' futures and the lifelong careers and research that many

scientists contribute to. The losses that the United States is currently facing due to the loss of billions of dollars from the NSF and its very funding to support and fuel many projects all across the United States that will ultimately affect and impact the future of the united states to, accordingly raising and question the future of the country. The National Science Foundation, also known as the NSF, is one of the largest and most influential federal agencies that is responsible for disbursing funding for research and education in science, technology, engineering, mathematics, and social science fields. Due to the large influence from the NSF grants and how they support a wide range of both academic and scientific work, these terminations of these grants can affect anyone and everyone even those who are not directly linked to them because these grants have effects on all fields and resources in the united states due to the termination of the oppurtities in research and education that was lost due to these grant terminations from the trump administration.

This paper will analyze the data set "National Science Foundation Grant Terminations Under the Trump Administration." This dataset was made by Grant Watch and imported into a CSV file by Noam Ross and Jhonthegeek. This data set focuses on the illegal grant terminations for the NSF by the trump administration. This data has (n=1041) observations. This data also has twenty-one variables; for more information on the variables, a data dictionary is provided below this paragraph that is from the tidyverse website. These variables did not include whether the organizations were private or public, so I did some research on which organizations in the data set were private or Public organizations, which creates the new dataset having twenty-two variables, in order to explore all my hypotheses that I had researched for this dataset. The overall goal of this analysis is to be able to identify which areas are most affected and to have a better understanding of how this type of data shows patterns of federal research funding losses and why they were canceled. As we explore this analysis, it will include Tables, Visualizations, and tests of hypotheses for this data.

Figure 1

Data Dictionary (Tidyverse/Tidytuesday/Github)

variable	class	description
grant_number	character	The numeric ID of the grant.
project_title	character	The title of the project the grant funds.
termination_letter_date	date	The date a termination letter was received by the organization.
org_name	character	The name of the organization or institution funded to do the project.
org_city	character	The name of the organization's city.
org_state	character	The organization's two-letter state abbreviation.
org_district	character	The congressional district (state and number) where the organization is located.
usaspending_obligated	double	The amount of money, via USAspending.gov, that NSF had committed to funding.
award_type	character	The type of grant.
directorate_abbrev	character	The three-letter abbreviation of the NSF directorate name.
directorate	character	The NSF directorate (the highest level of organization), which administered the grant.
division	character	The NSF division (housed within directorate) which administered the grant.
nsf_program_name	character	The name of the funding program under which the grant was made.
nsf_url	character	The URL pointing to the award information in the public NSF award database.
usaspending_url	character	The URL pointing to budget and spending information at the public USAspending.gov website.
nsf_startdate	date	The start date of the project.
nsf_expected_end_date	date	The date the project was expected to end.
org_zip	character	The 5- or 9-digit ZIP code of the organization receiving the grant.
org_uei	character	The unique entitity identifier (UEI) of the organization recieving the grant, used across U.S. government databases.
abstract	character	The text of the project abstract, describing the work to be done.
in_cruz_list	logical	Whether the project was in a list of NSF projects named by U.S. Senator Ted Cruz that he claimed "promoted Diversity, Equity, and Inclusion (DEI) or advanced neo-Marxist class warfare propaganda."

Note. The main variables I had used in the dataset were org_name, org,_state, usaspending_obligated, and directorate_abbrev.

Method

Variables used

In the dataset “National Science Foundation Grant Terminations Under the Trump Administration, " The variables I focused on the most to create questions and hypotheses were the org_name, org,_state, usaspending_obligated, Privatepublic, and directorate_abbrev. Where the

directorate_abbrev was used to examine the NSF directorates that had the highest amounts of grant terminations. The variable org_state was used to compare the grant terminations and their effects state by state and on a geographical level too. The Variable usaspending_obligated was used to consider the amount of money that was lost for the funding of each grant that was terminated by the Trump Administration and the NSF. The org_names were used to help create the new variable Private public to compare the damages on private and public organizations due to the grant terminations.

Questions & Hypotheses

This project had focused on 5 Main questions and their hypotheses, which are according to them. These will be listed out below in a question, and 2 hypotheses will be provided along with these questions. 1) In which directorate had the most grants terminated? H1: It was the education grants that were terminated the most. H0: Education grants are the least likely to be affected. 2) Which states and regions were most affected or had the most abundant amounts of grant terminations? H1: I believe that California would be one of the top states most affected, and the East Coast would be the most affected region. H0: The East Coast region and California were not affected much at all by the grant terminations. 3) Where were Private or Public organizations most affected or impacted by these grant terminations? H1: Yes, public organizations had suffered the most from grant terminations. H2: Yes, the private organizations had suffered the most from grant terminations.H0: Neither public nor private organizations had suffered the most from the grant terminations. 4) Which of the states surrounding and including Illinois has the most grants terminate out of the states surrounding Illinois? H1: I believe Illinois would have the most grants terminated out of the states surrounding Illinois. H2: Any of the other states surrounding Illinois were affected the most by grant terminations. H0: All the states are effectively equal in their effects due to the grant terminations. 5) Assuming California would be number one, which was most

affected across all the states, what would be the difference between the amounts of grants terminated and the money that was lost to the terminations across the whole United States? H1: California had a significant difference. H0: There was no significant difference.

Data Analysis

The data was analyzed by coding using R. I used the packages ggplot2,dplyr, here, maps, scales, tidyr, readr, gt, and webshot2. To create, manipulate, clean, test, and ect the results of my research on this data set. By using all these packages I have tested, I have created both tables and graphs to present my findings to whoever is reading this paper.

Statistical Testing

While exploring my data, I did many tests of my hypotheses, like ANOVA, Chi-square, Tukey Post Hoc, Kruskal-Wallis test, Wilcox Rank Sum test, and T-tests. To gather further information on my hypotheses and the results of what my data has offered to me. These tests were chosen because of the main variables being compared to each other, and I picked out the most interesting results from these tests to highlight in this paper. These tests were also used with tables and visualizations to identify any patterns in my data.

Results

Which directorate had the most grants terminated?

Table 1

Title: The top five directorates that had been affected the most across the whole data set

Top 5 NSF Directorates by Grant Terminations

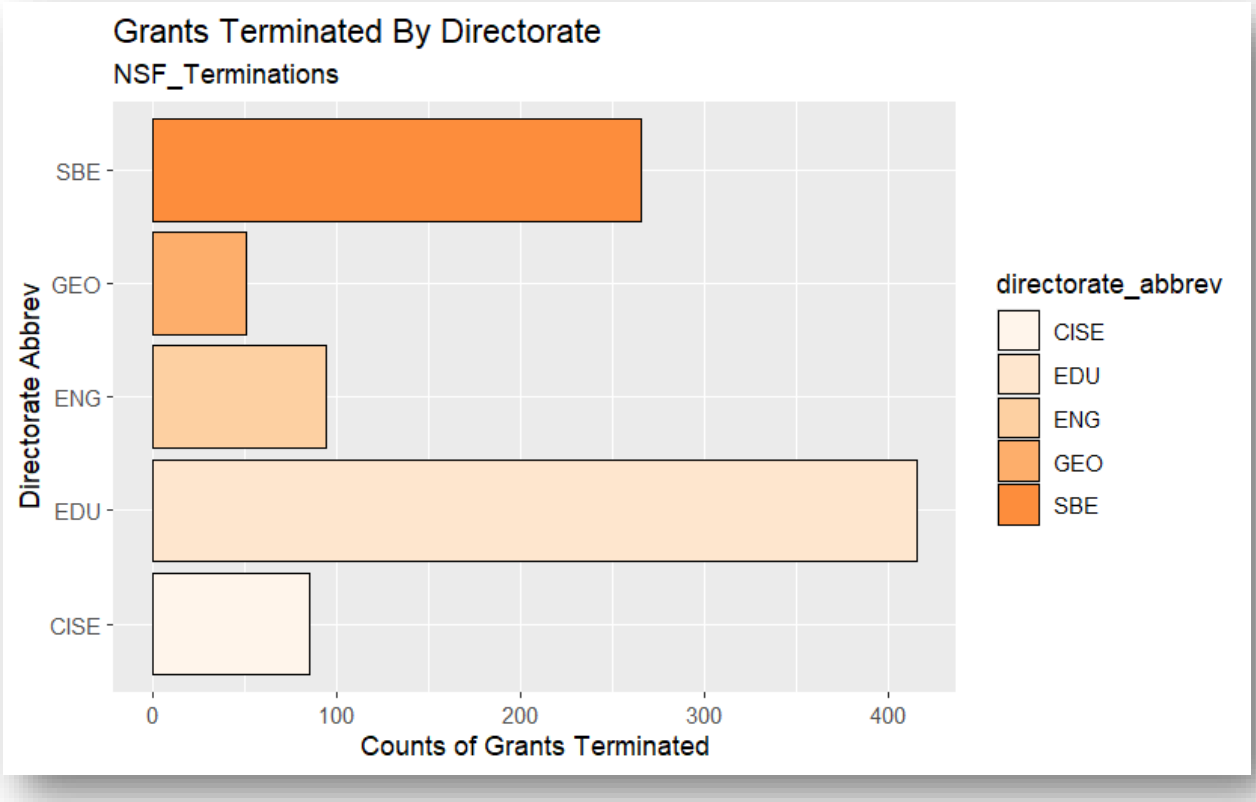
Count of terminated NSF grants by directorate

Directorate	Number of Terminated Grants
EDU	416
SBE	266
ENG	94
CISE	85
GEO	51

Note. This was created in R, and these are all the states included, and it comes from the counts of the grants terminated and the amounts of money lost by each state. Abbreviation meanings**.** EDU = Education; SBE = Social, Behavioral, and Economic Sciences; ENG = Engineering; GEO = Geosciences.

Figure 1

Title: Grants Terminated by Directorate



Note. Abbreviation meaning**.** EDU = Education; SBE = Social, Behavioral, and Economic Sciences; ENG = Engineering; GEO = Geosciences.

Test results

The results of the Kruskal-Wallis test were conducted to determine whether the amount of NSF funding lost differed across the top five directorates. The results showed a statistically significant difference in funding loss by directorate, $X^2(4) = 149.36$, $p < .001$. This means that the distribution of obligated funding lost was not equal across directorates. In other words, some NSF directorates experienced much larger funding losses than others.

Which states and regions were most affected or had the most abundant amounts of grant terminations?

Where were private or Public organizations where most affected or impacted by these grant terminations?

Which of the states surrounding and including Illinois have the most grants terminated out of the states surrounding Illinois?

Assuming California would be number one, which was most affected across all the states, what would be the difference in the amounts of grants terminated and the money that was lost to the terminations across the whole United States?

Conclusion

In conclusion, this paper has examined the NSF Grant termination under the Trump administration in 2025. By looking at how the funding, losses were distributed across the states, districts, California vs non-California grants, private and public, and more. All these results show the overall effects of these grants that were ultimately not distributed. One of the most significant findings from the comparison of NSF doctoral programs where the ANOVA test showed that the average amount of money lost differed across each of the research areas. This ultimately suggests that certain fields of research were tattered by these grants and were significantly affected more than others. The results from the California comparison show a more complex result. That showed my assumptions of a greater difference were actually not that significant at all for the Test and the Wilcoxon test. That ultimately meant that Californian grants did not lose significantly more money on average than the grants in other states. Overall, these findings show that NSF Grant terminations had unequal impacts that depend on how the data was examined, as noted that this data is biased, like its roots here, has a negative opinion of the trump administration. By looking at the money lost, it gives one part of the start, while looking at the counts, it gives another, but when combining all of these factors, it gives a fuller picture of the reasoning and the biases toward the trump administration. These results suggest that the terminations that took place had a large effect

on the entirety of the United States and prove how these grants affect the United States and shape our public institutions and possible well-being across the United States due to this major loss. The final thing I would like to add is that the issue is continuing, and I could do better in the future if I make a CSV of the 2026 termination to combine with the 2025 terminations.

References

Bork, F. (2026, April 19). National Science Foundation's future in limbo as Trump eyes cuts. Retrieved April 23, 2026, from The Hill website: https://thehill.com/homenews/administration/5835619-future-nsf-research-limbo/

Duncan, M. (2026, February 20). What Gets Lost When Trump Removes Black History From Historic Sites. Retrieved from Architectural Digest website: https://www.architecturaldigest.com/story/what-gets-lost-when-trump-removes-black-history-from-buildings

Ellis, N. T. (2025, January 23). What is DEI, and why is it dividing America? Retrieved from CNN website: https://www.cnn.com/2025/01/22/us/dei-diversity-equity-inclusion-explained

Grant Witness.
(2025). Retrieved from Grant Witness website: https://grant-witness.us/

Miller, K., & Zimmer, C. (2025, April 22). National Science Foundation Terminates Hundreds of Active Research Awards. *The New York Times*. Retrieved from https://www.nytimes.com/2025/04/22/science/trump-national-science-foundation-grants.html

rfordatascience.
(2025). tidytuesday/data/2025/2025-05-06/readme.md at main · rfordatascience/tidytuesday. Retrieved April 23, 2026, from GitHub website: https://github.com/rfordatascience/tidytuesday/blob/main/data/2025/2025-05-06/readme.md

Schwabish, J., & Axelrod, J. (2025, July 9). NSF Has Canceled More Than 1,500 Grants. Nearly 90 Percent Were Related to DEI. Retrieved from Urban Institute website: https://www.urban.org/urban-wire/nsf-has-canceled-more-1500-grants-nearly-90-percent-were-related-dei

rfordatascience/tidytuesday. Retrieved April 23, 2026, from GitHub website: https://github.com/rfordatascience/tidytuesday/blob/main/data/2025/2025-05-06/readme.md